

STRATEGY VERIFICATION & DEPLOYMENT ASSESSMENT

MarketMate Deep Validation & Portfolio Attribution

Comprehensive 12-year deep validation of all MarketMate trading strategies, revealing that only the 8-Gate SMC pipeline holds a robust edge while the Monday Gap Fade and eight other strategies fail conclusively out-of-sample.

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MarketMate Research Division

DEEP VALIDATION REPORT

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1. Executive Summary

The deep validation of all MarketMate strategies reveals a sobering picture. Only the 8-Gate SMC pipeline shows a robust, validated edge. The Monday Gap Fade (MM-002), which appeared profitable on short-window H4 data (PF 1.17), fails conclusively on 12-year daily data with a portfolio-wide PF of 0.65. Eight of twelve tested strategies are clear rejects. The project must transition from strategy creation to validation, attribution, and portfolio construction.

Key Findings

- MM-002 FAILS deep validation: PF 0.65, Bootstrap $P(PF < 1) = 100\%$, no profitable year in 12 years
- 8-Gate SMC is the only validated edge: PF 3.65, Sharpe 10.44 (limited to 2-3 years H4 data)
- MM-012 (Gap Fill) is marginally profitable (PF 1.23) but trades the same alpha as MM-002
- MM-005 (RSI Divergence) shows PF 2.0 but only 6 trades - statistically meaningless
- 8 strategies are confirmed rejects and archived in the Strategy Graveyard
- Regime analysis reveals MM-002 works ONLY in RANGING/LOW_VOL conditions
- The risk engine already exists as a unified 6-module decision pipeline

2. MM-002 Deep Validation Results

2.1 Portfolio Statistics (12-Year Daily Data, 18 Instruments)

Metric	Value
Total Trades	3,587
Profit Factor	0.6459
Win Rate	56.2%
Average R	-0.187
Max Drawdown	696.1R

Table 1: MM-002 Portfolio-Level Statistics

2.2 Year-by-Year Performance

Year	Trades	PF	WR	Avg R
2014	182	0.963	65.9%	-0.0149
2015	237	0.282	48.9%	-0.8732
2016	239	0.301	55.2%	-0.7930
2017	253	0.856	59.7%	-0.0620
2018	308	0.737	55.2%	-0.1142
2019	321	0.871	59.8%	-0.0504
2020	352	0.626	49.4%	-0.1887
2021	308	0.639	51.6%	-0.1703
2022	294	0.710	55.4%	-0.1274
2023	288	0.735	54.2%	-0.1155
2024	294	0.835	58.5%	-0.0677
2025	326	0.994	61.0%	-0.0023
2026	185	1.083	60.5%	+0.0317

Table 2: MM-002 Year-by-Year Performance (2014-2026)

2.3 Walk-Forward Validation (5-Fold)

Fold	Train PF	Test PF	Degradation
1	0.365	0.753	2.06x
2	0.753	0.734	0.98x
3	0.734	0.627	0.85x
4	0.627	0.836	1.33x
5	0.836	0.939	1.12x

Table 3: Walk-Forward Validation Results

Walk-forward conclusion: All out-of-sample folds show $PF < 1.0$. The strategy is consistently unprofitable out-of-sample.

2.4 Monte Carlo Simulation (10,000 Simulations, 1% Risk)

Metric	Value
P(Ruin)	0.00%
Mean Final Balance	\$0.92 (from \$1,000 start)
P95 Max DD	99.9%
Still Profitable	0.0%

Table 4: Monte Carlo Simulation Results

2.5 Bootstrap Monte Carlo (10,000 Resamples)

Metric	Value
PF Mean	0.6464
PF Median	0.6457
PF 5th-95th CI	[0.6018, 0.6932]
P(PF < 1.0)	100.0%
Avg R Mean	-0.1872
Avg R CI	[-0.2209, -0.1542]

Table 5: Bootstrap Monte Carlo Results

2.6 Trade Removal Test

Scenario	PF	P(PF<1)
Base (3,587 trades)	0.6459	100%
Remove 10% winners	0.5816	100%
Remove 20% winners	0.5168	100%
Remove 30% winners	0.4525	100%

Table 6: Trade Removal Sensitivity Test

2.7 Gap Direction Analysis

Direction	Trades	PF	WR	Avg R
Gap Up (short)	1,903	0.574	53.9%	-0.238
Gap Down (long)	1,684	0.738	58.9%	-0.130

Table 7: Gap Direction Analysis

2.8 Spread/Slippage Stress (NZDUSD)

Multiplier	Spread PF	Slippage PF
1x	1.35	1.35
2x	1.28	1.34
3x	1.20	-
5x	1.07	1.29
10x	-	1.22

Table 8: Spread/Slippage Stress Test on NZDUSD

VERDICT: MM-002 FAILS deep validation. The edge is a statistical artifact of short data windows.

3. Instrument-Level Analysis

3.1 Top Instruments (Marginally Profitable)

Symbol	Trades	PF	WR	Avg R	MDD R
NZDUSD	174	1.23	66.1%	+0.0724	5.1
USDCHF	155	1.20	67.1%	+0.0622	8.3
EURJPY	164	1.10	60.4%	+0.0355	9.2
GBPUSD	157	1.03	61.2%	+0.0120	12.7
AUDUSD	166	1.01	60.8%	+0.0055	11.2

Table 9: Top Instruments for MM-002

3.2 Worst Instruments

Symbol	Trades	PF	WR	Avg R	MDD R
BTCUSD	366	0.15	42.6%	-1.322	485.2
US30	123	0.42	43.1%	-0.321	40.6
US500	176	0.62	52.3%	-0.180	40.4
XAGUSD	316	0.73	52.8%	-0.124	42.4
XAUUSD	235	0.76	57.0%	-0.104	31.9

Table 10: Worst Instruments for MM-002

BTCUSD is the single worst performer, with a catastrophic PF of 0.15 and a maximum drawdown of 485.2R. Crypto instruments are fundamentally incompatible with gap-fade strategies due to their trending nature and extreme volatility.

4. Regime-Conditional Analysis

Key finding: MM-002's performance is entirely regime-dependent. Gap fade strategies work in RANGING/LOW_VOL conditions and fail in TRENDING/HIGH_VOL.

4.1 EURUSD by Regime

Regime	Trades	PF	WR	Avg R
RANGING	12	3.29	83.3%	+0.389
MIXED	22	1.66	68.2%	+0.215
LOW_VOL	43	1.16	67.4%	+0.051
HIGH_VOL	60	0.75	51.7%	-0.117
TRENDING	30	0.62	53.3%	-0.165

Table 11: EURUSD Regime-Conditional Performance

4.2 GBPUSD by Regime

Regime	Trades	PF	WR	Avg R
MIXED	13	7.15	92.3%	+0.478
RANGING	15	1.31	66.7%	+0.105
HIGH_VOL	48	0.98	58.3%	-0.009
TRENDING	35	0.89	54.3%	-0.041
LOW_VOL	43	0.76	58.1%	-0.098

Table 12: GBPUSD Regime-Conditional Performance

Conclusion: Gap fade strategies work in RANGING/LOW_VOL and fail in TRENDING/HIGH_VOL. This is the single most important finding for regime-aware deployment.

5. Portfolio Attribution

5.1 Alpha Sources

Strategy	Alpha Source	PnL (R)	Contribution
MM-002	Weekend gap mean reversion	-670.63	Dominant loser
MM-012	Weekend gap fill	+5.15	Marginal
SMC-8G	Structural liquidity + trend	Not tested in daily	Core system

Table 13: Alpha Source Attribution

5.2 Correlation Analysis

Correlation between MM-002 and MM-012: **0.063** (low monthly correlation but same failure mode: trending markets).

Rolling correlation (MM-002 vs MM-012): Trending upward from **0.09 to 0.30** over recent months, indicating convergence.

Key insight: MM-002 and MM-012 trade the same phenomenon (weekend gap). Low correlation is misleading - they share the same failure mode.

6. Strategy Graveyard

9 strategies archived with documented rejection reasons:

Strategy	Name	PF	Trades	Verdict
MM-002	Monday Gap Fade	0.65	3587	FAILED_DEEP_VALIDATION
MM-003	Key Level Defensive Fade	0.55	110	REJECT
MM-009	Liquidity Sweep Quick Reversal	0.84	46	MARGINAL_REJECT
MM-019	Order Block Retest	0.54	83	REJECT (SMC filter)
MM-020	Fair Value Gap Fill	0.53	176	REJECT (SMC filter)
MM-021	Market Structure Shift	0.57	196	REJECT (SMC filter)
MM-022	Session Raid + Reversal	0.00	0	REJECT (zero signals)
MM-023	London Breakout	0.70	372	REJECT
MM-024	NY Reversal	0.11	31	REJECT (catastrophic)
MM-025	ATR Compression Breakout v2	0.45	208	REJECT

Table 14: Strategy Graveyard - Archived Strategies

7. Surviving Deployment Candidates

7.1 8-Gate SMC Pipeline

Metric	Value
PF	3.65
Sharpe	10.44
WR	67%
Trades	291
Verdict	Level 3 PAPER TRADE

Table 15: 8-Gate SMC Pipeline Performance

Critical Risk: H4 data limited to 2-3 years. LTF confirmation only ~60 days. All bias checks PASS except data limitation warning.

7.2 MM-005 (RSI Divergence)

PF: 2.0 | WR: 67% | Trades: 6

Verdict: INSUFFICIENT SAMPLE - do NOT deploy until 30+ trades accumulated.

7.3 MM-012 (Gap Fill Weekend)

PF: 1.23 | WR: 55% | Trades: 134 | Avg R: 0.10

Verdict: MARGINAL - same alpha as MM-002, only viable on NAS100/US30/GBPUSD.

8. Risk Engine Status

The unified risk engine already exists at `signal_engine/strategies/risk_manager.py` with 6 modules operating as a single decision pipeline:

Module	Name	Function
MM-004	Drawdown Circuit Breaker	Daily/weekly/monthly limits
MM-006	Pre-Weekend Protocol	Thursday close, Friday/Sunday blocks
MM-007	Thin Market Guard	Spread, ATR, blocked hours
MM-008	Consecutive Loss Protocol	Strategy + portfolio level
MM-010	Correlation Cap	Position count + pairwise correlation
MM-011	Equity Curve Brake	SMA deviation + 20% halt

Table 16: Risk Engine Modules

Usage: `rm.check_signal(signal, spread, atr, ...)` returns (allowed, reason, size_multiplier)

9. Recommendations

1. **STOP creating new strategies.** The validation data is clear: only SMC-based approaches work.
2. **DEPLOY 8-Gate SMC** to 90-day paper trading with NO parameter changes.
3. **BUILD a regime detection module** and test MM-002 with RANGING-only filter on NZDUSD/USDCHEF.
4. **ARCHIVE all graveyard strategies permanently.** Do not delete - prevent rediscovery.
5. **VALIDATE 8-Gate SMC** on longer data windows if possible (use daily data for structural analysis).
6. **TRACK regime state** in paper trading to build a live regime-performance database.
7. **REVISIT MM-009** after paper trading confirms ETHUSD liquidity sweep edge.